Q1. JEE Main 2026 (21 January Shift 2)

A particle having electric charge 3×10^{-19}C and mass 6×10^{-27} kg is accelerated by applying an electric potential of 1.21 V. Wavelength of the matter wave associated with the particle is $\alpha \times 10^{-12}$ m. The value of α is ____ -
(Take Planck's constant $= 6.6 \times 10^{-34}$ J. s)

Q2. JEE Main 2026 (23 January Shift 1)

The de Broglie wavelength of an oxygen molecule at 27°C is $x \times 10^{-12}$ m. The value of x is (take Planck's constant $= 6.63 \times 10^{-34}$ J. s, Boltzmann constant $= 1.38 \times 10^{-23}$ J/K, mass of oxygen molecule $= 5.31 \times 10^{-26}$ kg)

(1) 24 (2) 26 (3) 30 (4) 20

Q3. JEE Main 2026 (28 January Shift 2)

Number of photons of equal energy emitted per second by a 6 mW laser source operating at 663 nm is ____.
(Given : $\mathrm{h} = 6.63 \times 10^{-34}$ J. s and $\mathrm{c} = 3 \times 10^{8}$ m/s)

(1) 2×10^{16} (2) 10×10^{15}
(3) 5×10^{15} (4) 5×10^{16}

Q4. JEE Main 2026 (24 January Shift 2)

When a light of a given wavelength falls on a metallic surface the stopping potential for photoelectrons is 3.2 V. If a second light having wavelength twice of first light is used, the stopping potential drops to 0.7 V. The wavelength of first light is ____ m. $\left(\mathrm{h} = 6.63 \times 10^{-34}\ \mathrm{J.s}, \mathrm{e} = 1.6 \times 10^{-19}\mathrm{C}, \mathrm{c} = 3 \times 10^{8}\ \mathrm{m/s}\right)$

(1) 2.5×10^{-7} (2) 2.2×10^{-8}
(3) 3.1×10^{-7} (4) 2.9×10^{-8}

Q5. JEE Main 2026 (22 January Shift 2)

Light is incident on a metallic plate having work function 110×10^{-20} J. If the produced photoelectrons have zero kinetic energy then the angular frequency of the incident light is ____ rad/s. $\left(\mathrm{h} = 6.63 \times 10^{-34}\ \mathrm{J.s}\right)$.

(1) 1.04×10^{13} (2) 1.04×10^{16}
(3) 1.66×10^{16} (4) 1.66×10^{15}

Q6. JEE Main 2026 (21 January Shift 1)

A light wave described by $E = 60\left[\sin\left(3 \times 10^{15}\right)t + \sin\left(12 \times 10^{15}\right)t\right]$ (in SI units) falls on a metal surface of work function 2.8 eV. The maximum kinetic energy of ejected photoelectron is (approximately) ____ eV.
$\left(h = 6.6 \times 10^{-34}\ \mathrm{J.\ s.}\ \text{and}\ e = 1.6 \times 10^{-19}\mathrm{C}\right)$

(1) 7.8 (2) 6.0 (3) 5.1 (4) 3.8

Q7. JEE Main 2026 (28 January Shift 1)

The ratio of de Broglie wavelength of a deutron with kinetic energy E to that of an alpha particle with kinetic energy $2E$, is $n : 1$. The value of n is ____.
(Assume mass of proton = mass of neutron) :

ANSWERS AND SOLUTIONS

1. 10 **2.** (2) **3.** (1) **4.** (1) **5.** (2) **6.** (3) **7.** 2

1. **(10)** The de Broglie wavelength is $\lambda = \frac{h}{mv}$ where v is found from energy conservation.

From $qV = \frac{1}{2}mv^2$: $v = \sqrt{\frac{2qV}{m}} = \sqrt{\frac{2\times3\times10^{-19}\times1.21}{6\times10^{-27}}} = \sqrt{1.21\times10^8} = 1.1\times10^4$ m/s.

Therefore: $\lambda = \frac{6.6\times10^{-34}}{6\times10^{-27}\times1.1\times10^4} = \frac{6.6\times10^{-34}}{6.6\times10^{-23}} = 10^{-11}$ m = 10×10^{-12} m.

Thus $\alpha = 10$.

2. **(2)** The de Broglie wavelength is $\lambda = \frac{h}{mv}$. For an ideal gas, $v_{rms} = \sqrt{\frac{3kT}{m}}$

Therefore: $\lambda = \frac{h}{\sqrt{3mkT}}$

Substituting values: $\lambda = \frac{6.63\times10^{-34}}{\sqrt{3\times5.31\times10^{-26}\times1.38\times10^{-23}\times300}}$

$\lambda = \frac{6.63\times10^{-34}}{\sqrt{6.62\times10^{-46}}} = \frac{6.63\times10^{-34}}{2.57\times10^{-23}}$

$\lambda = 2.58\times10^{-11}$ m $= 25.8\times10^{-12}$ m, so $x \approx 26$

3. **(1)** Power of laser: $P = 6$ mW $= 6\times10^{-3}$ J/s

Energy of a single photon at wavelength $\lambda = 663$ nm:

$E = h\nu = \frac{hc}{\lambda} = \frac{6.63\times10^{-34}\times3\times10^8}{663\times10^{-9}}$

$E = \frac{6.63\times3}{663}\times10^{-34+8+9} = \frac{19.89}{663}\times10^{-17} = 3\times10^{-19}$ J

Number of photons emitted per second:

$N = \frac{\text{Total Energy per second}}{\text{Energy per photon}} = \frac{6\times10^{-3}}{3\times10^{-19}} = 2\times10^{16}$

4. **(1)** Using Einstein's photoelectric equation with stopping potentials:

For first light, $h\nu_1 = \phi + eV_{s1}$.

For second light with $\lambda_2 = 2\lambda_1$ (so $\nu_2 = \nu_1/2$): $h\nu_1/2 = \phi + eV_{s2}$.

From given values: $h\nu_1 - 3.2e = h\nu_1/2 - 0.7e$ gives $h\nu_1/2 = 2.5e$.

Thus $\lambda_1 = \frac{hc}{5e} = \frac{6.63\times10^{-34}\times3\times10^8}{5\times1.6\times10^{-19}} \approx 2.5\times10^{-7}$ m.

5. **(2)** From Einstein's photoelectric equation, $h\nu = W + KE$.

With zero kinetic energy, $h\nu = W$.

The frequency is $\nu = \frac{W}{h} = \frac{110\times10^{-20}}{6.63\times10^{-34}} = 16.59\times10^{14}$ Hz.

The angular frequency is $\omega = 2\pi\nu = 2\pi\times16.59\times10^{14}$

$= 104.25\times10^{14} = 1.04\times10^{16}$ rad/s.

6. **(3)** Light contains two frequencies: $\omega_1 = 3\times10^{15}$ rad/s and $\omega_2 = 12\times10^{15}$ rad/s

For maximum KE, use higher frequency ω_2.

Energy: $E_2 = \frac{h\omega_2}{2\pi} = \frac{6.6\times10^{-34}\times12\times10^{15}}{2\pi}$

$= \frac{79.2\times10^{-19}}{6.28} = 12.6 \times 10^{-19}$ J $= 7.9$ eV

$KE_{max} = E_2 - \phi = 7.9 - 2.8 = 5.1$ eV

7. **(2)** The de Broglie wavelength is $\lambda = \frac{h}{\sqrt{2mE}}$.

For a deuteron (mass $m_d = 2m_p$) with energy E: $\lambda_d = \frac{h}{\sqrt{2\cdot 2m_p\cdot E}} = \frac{h}{\sqrt{4m_pE}}$

For an alpha particle (mass $m_\alpha = 4m_p$) with energy $2E$: $\lambda_\alpha = \frac{h}{\sqrt{2\cdot 4m_p\cdot 2E}} = \frac{h}{\sqrt{16m_pE}}$

The ratio: $\frac{\lambda_d}{\lambda_\alpha} = \frac{\sqrt{16m_pE}}{\sqrt{4m_pE}} = \sqrt{4} = 2$